

SHUBHIKA SHREE

Python Backend & Cloud Automation Engineer

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PROFESSIONAL SUMMARY

Software Engineer with 2 years of experience building automated backend systems and secure, multi-cloud infrastructure across Azure and AWS. Hands-on expertise in REST API development (Flask, FastAPI), Python-based cloud automation and workflow orchestration, and identity/secrets management (HashiCorp Vault, Azure Key Vault, OAuth 2.0). Currently pursuing an M.Tech in Artificial Intelligence & Data Science at IIT Patna.

TECHNICAL SKILLS

- **Programming & Backend:** Python, SQL, Flask, FastAPI, REST APIs, Redis
- **Cloud & Infrastructure:** Microsoft Azure, AWS, GCP, Azure Arc / Azure Local, Azure Resource Manager
- **Security & Identity:** HashiCorp Vault, Azure Key Vault, OAuth 2.0, SSH (Paramiko)
- **DevOps:** Git, CI/CD, Jenkins, ArgoCD, Kubernetes
- **Data & Analytics:** Pandas, Matplotlib, Seaborn, Streamlit, Excel
- **Testing & Tools:** Unit Testing, Behavior-Driven Development (BDD), JIRA

PROFESSIONAL EXPERIENCE

Relevance Lab Pvt Ltd

Nov 2024 – Present

Harmonize Provisioning | Fortune 50 Healthcare Client

Developed and maintained features across **24 Python Flask microservices** powering Harmonize, an infrastructure automation platform that orchestrates the end-to-end VM lifecycle (provisioning, configuration, validation, and decommissioning) across **AWS and Azure**, processing **5,000+ provisioning requests monthly** through a **150+ component state machine orchestration engine**.

Software Engineer

Apr 2026 – Present

- Engineered **Azure Arc / Azure Local** support across 4 microservices, integrating Azure Resource Manager APIs to enable unified VM lifecycle operations and metadata retrieval.
- Implemented automated tag/lock management and **Rubrik-based backup** automation, validation, and decommissioning workflows across Global and China regions.
- Automated Azure Local VM provisioning for Windows and Linux by enhancing bootstrap scripts with IMDS-based VM detection and metadata/tag extraction, normalizing provider configuration across **100+ virtual machines**.
- Integrated internal provisioning and database workflows with automated OS-to-database role mapping, reducing manual provisioning effort while maintaining compatibility with standard Azure VM workflows.
- Hardened Tufin failure handling in decommission workflows with automated API retries and exponential backoff, incident ticket creation, and targeted alert routing, reducing downtime by up to **~10 hours** per transient Tufin API outage (incl. China region).
- Modernized SSH connectivity across Harmonize microservices by replacing raw Paramiko connections with an internal SSH library using **HashiCorp Vault-signed ephemeral keys** and Redis caching (36-hour TTL), eliminating repeated Vault signing calls and improving reliability under concurrent production workloads.
- Migrated Basic Authentication to **OAuth 2.0 Client Credentials flow** across Azure, AWS, and GCP integrations with secure token acquisition, caching, and auto-refresh, eliminating static credential exposure in Global and China environments.
- Diagnosed and resolved **1,000+ production and decommissioning issues**.

Trainee Software Engineer

Nov 2024 – Mar 2026

- Contributed to automation and security hardening of the Harmonize platform, a multi-cloud infrastructure provisioning system supporting Azure and AWS environments.
- Strengthened infrastructure security by replacing static SSH credentials with **HashiCorp Vault-generated dynamic keys** for Linux VM authentication, eliminating long-lived SSH keys.
- Integrated **Azure Key Vault across 24 microservices** with proxy services for runtime secret retrieval, enabling centralized credential management and seamless secret rotation.
- Automated VM provisioning and decommissioning workflows across Azure and AWS, reducing manual intervention in lifecycle management.
- Automated Tableau VM zone placement using **IMDS API** integration and environment tag detection, eliminating manual zone configuration during provisioning.

PROJECTS

SPPU Result Converter | Python, Pandas, Streamlit, Regex, Excel

- Built a **Streamlit-based web application** that converts unstructured university result PDFs into structured Excel reports using regex-driven extraction and automated formatting pipelines.
- Reduced manual processing time from **~3 days to under 10 minutes (~99% reduction)**; adopted operationally by faculty at D.Y. Patil College of Engineering.

Superstore Sales Data Analytics | Python, Pandas, Matplotlib, Seaborn

- Conducted end-to-end sales data analysis across regions and product categories to identify trends and top-performing segments.
- Performed data cleaning, exploratory analysis, and visualization to translate raw sales data into actionable business insights.

EDUCATION

M.Tech in Artificial Intelligence & Data Science — Indian Institute of Technology (IIT) Patna | CGPA: 8.56

2025 – Present

B.E. in Computer Science Engineering — D.Y. Patil College of Engineering, Pune | CGPA: 8.6

2020 – 2024

ACHIEVEMENTS

- Recognized by leadership for dedication and technical contribution during critical Rubrik testing and release phases.
- Received formal appreciation from the client's Cloud team for professional commitment and consistent delivery quality.